Frontier



NuWa Solutions

R&D application

2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

NuWa Solutions

Organization location (we welcome applicants from anywhere in the world)

Maine, U.S.A.

Name(s) of primary point(s) of contact for this application

Dr. Elliott Roberts

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

For-profit marine carbon removal company developing deployment-ready ocean and inland water alkalinity enhancement pathways for durable atmospheric CO_2 removal.

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

Constraining $Ca(OH)_2$ precipitation and alkalinity retention in wastewater and coastal discharge settings

[1] We use “project” throughout this template, but the term is not intended to denote a single facility or experimental setup. The “project” being proposed to Frontier could include multiple facilities/locations for research activities.

b. **Project summary and field impact:** Provide an overview of your core research goal and the benefits you think it could provide to the broader CDR field. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

Ocean alkalinity enhancement and inland water alkalinity enhancement are promising durable carbon removal pathways[1,2], but both remain limited by uncertainty around secondary precipitation during alkalinity addition[3]. This uncertainty is especially important in wastewater and coastal discharge settings, where feedstock delivery, wastewater chemistry, receiving-water chemistry, and early dilution conditions determine whether $Ca(OH)_2$ remains as persistent alkalinity or is rapidly lost through mineral formation.

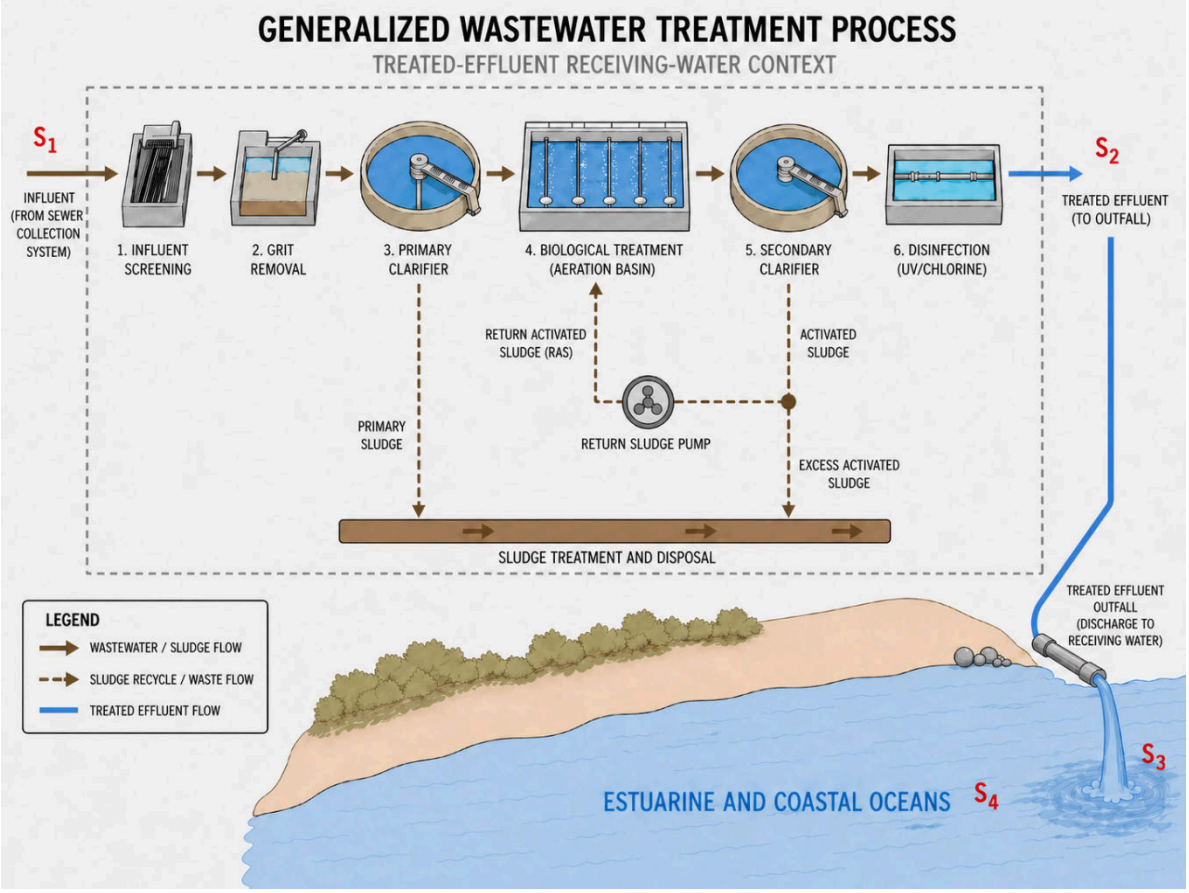


Figure 1. Generalized wastewater treatment process and treated-effluent receiving-water context, showing indicative sample collection points (S_1–S_4) used to characterize plant-side chemistry and receiving-water conditions prior to laboratory-controlled $Ca(OH)_2$ perturbation experiments. S_1 represents influent or early-stage wastewater entering the treatment process, S_2 represents final treated effluent prior to discharge, S_3 represents the outfall discharge or immediate near-field receiving-water zone, which may correspond to an above-ground, shoreline, nearshore, or submarine

discharge configuration depending on site infrastructure, and S_4 represents background receiving water in the adjacent estuarine or coastal environment. Final sample locations may vary depending on plant configuration, access, logistics, safety requirements, and partner approvals.

NuWa Solutions proposes a focused experimental program to quantify $Ca(OH)_2$ behavior in wastewater effluents and adjacent receiving waters under deployment-relevant conditions. The project is anchored in a temperate municipal wastewater and coastal receiving-water context, with comparator matrices representing warm semi-enclosed coastal and tropical estuarine conditions where feasible. The goal is to define a transferable chemistry and infrastructure envelope relevant to wastewater-enabled and coastal alkalinity enhancement, rather than to validate any specific deployment site. Specific sampling locations, partner facilities, and regional access details will be handled confidentially and will depend on logistics, safety requirements, partner approvals, and project timing.

The project places particular emphasis on wastewater-adjacent systems because the controlling chemistry begins within the wastewater itself, before alkalinity is added and before discharge into the receiving environment. Wastewater-associated ammonium, organic matter respiration, nutrient loading, and treatment-stage variability can alter total alkalinity (TA), dissolved inorganic carbon (DIC), and pH in ways that affect precipitation risk after $Ca(OH)_2$ addition. As a result, thresholds derived from simplified seawater experiments may not translate directly to realistic discharge settings.

The work will be carried out in collaboration with the Wei-Jun Cai Lab at the University of Delaware, which will lead the controlled carbonate chemistry experiments. NuWa will lead overall project integration, define the deployment-relevant experimental scenarios, and translate the results into implications for future pilot evaluation and open-system MRV. This division of roles is deliberate: the Cai Lab leads the chemistry, while NuWa ensures the project addresses the practical constraints that matter for future OAE and IWAE deployment.

The core objective is to determine the combinations of wastewater chemistry, receiving-water chemistry, delivery mode, and initial dilution conditions under which $Ca(OH)_2$ is retained efficiently rather than partially lost in the near field. This directly addresses Frontier's priority area on maximizing alkalinity addition rates for OAE and IWAE while also improving the empirical basis for near-field monitoring and interpretation. For this work to unlock step-change advancements in cost and scale, the field needs usable empirical constraints on when $Ca(OH)_2$ addition can be conducted efficiently without unacceptable alkalinity loss in the near field. If this project can define those constraints across representative wastewater and coastal receiving-water conditions, it would reduce reliance on overly conservative dosing assumptions, improve confidence in comparative evaluation of wastewater and coastal settings, and make infrastructure-linked deployment pathways more technically and economically credible. If successful, the project will strengthen future pilot evaluation, improve early-stage evaluation of candidate receiving-water environments, and support improved interpretation of dosing and monitoring strategies. More broadly, the resulting datasets and interpretation summary would help the CDR field move from generalized assumptions about $Ca(OH)_2$ behavior toward deployment-relevant scientific constraints that can accelerate pre-pilot evaluation in open-system alkalinity enhancement.

References

1. Renforth, P. & Henderson, G. Assessing ocean alkalinity for carbon sequestration. Rev. Geophys. 55, 636–674 (2017).
2. Cai, W.-J. & Jiao, N. Wastewater alkalinity addition as a novel approach for ocean negative carbon

emissions. The Innovation 3, 100272 (2022).
3. Moras, C. A., Bach, L. T., Cyronak, T., Joannes-Boyau, R. & Schulz, K. G. Ocean alkalinity enhancement – avoiding runaway $CaCO_3$ precipitation during quick and hydrated lime dissolution. Biogeosciences 19, 3537–3557 (2022).

c. **Project objectives:** Describe your project in more detail: what are your key research questions and proposed approach for answering them, including methods, project location, and scale where applicable? Tell us why your approach is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

The central challenge for $Ca(OH)_2$-based OAE is to determine under what wastewater and receiving-water conditions added alkalinity persists long enough to support efficient atmospheric CO_2 uptake. In wastewater and coastal discharge settings, the answer depends on the chemistry of the wastewater itself, the receiving-water carbonate system, the mode of feedstock delivery, and the short-timescale dilution conditions that control near-field carbonate system evolution.

This project will address that question through a controlled experimental campaign using wastewater effluents, receiving-water samples, and chemistry-matched analog systems spanning representative deployment conditions. The study is anchored in a temperate municipal wastewater and coastal receiving-water context, which provides the primary operational frame for the work. Comparator matrices representing warm semi-enclosed coastal and tropical estuarine systems will broaden the parameter space where access, timing, and partner approvals allow. These comparators are included to test transferable chemistry relationships, not to disclose specific deployment locations.

These environmental contexts were selected to span the wastewater and receiving-water conditions most likely to control $Ca(OH)_2$ performance in infrastructure-linked alkalinity enhancement. The primary temperate municipal wastewater and coastal receiving-water context provides a near-term frame for evaluating plant-side chemistry, treated-effluent variability, dilution behavior, and receiving-water response under regulated coastal discharge conditions. The warm semi-enclosed coastal comparator broadens the parameter space to include higher temperature, stronger wastewater influence, riverine inputs, stratification potential, restricted exchange, and rapid coastal development, all of which may alter carbonate-system evolution and shift secondary-precipitation thresholds relative to temperate outfall settings. A tropical estuarine comparator may be included if access and timing allow, adding a higher-freshwater-influence and biologically active receiving-water class to test whether retained alkalinity relationships remain predictable across broader TA/DIC, salinity, and organic-matter gradients. Prior synthesis also shows that wastewater effluent carbon levels vary with treatment technology and environmental setting, supporting a comparative design organized around wastewater and receiving-water archetypes rather than a single-site design[4].

The project will address five main questions.

1. Under what combinations of wastewater chemistry, receiving-water chemistry, and dilution does $Ca(OH)_2$ addition trigger measurable secondary precipitation?
2. How do feedstock preparation and delivery mode, including slurry-based addition, affect short-timescale alkalinity persistence?
3. Which variables best predict retained alkalinity efficiency in the near-field conditions?
4. Does wastewater-associated ammonium, organic matter, and nutrient loading systematically

shift precipitation thresholds relative to less impacted coastal waters?
5. How much do these relationships vary across temperate and tropical systems, or across different classes of treatment plant and receiving-water context?

The work will be led jointly by NuWa Solutions and the Wei-Jun Cai Lab at the University of Delaware. The Cai Lab will lead the experimental carbonate chemistry program, including controlled perturbation experiments and analytical measurement. NuWa will ensure that the experimental design remains deployment-relevant and will translate the results into implications for future pilot evaluation, monitoring, and MRV. That deployment-translation role is central to the project.

Experimentally, the study will evaluate $Ca(OH)_2$ addition in wastewater effluents and adjacent receiving waters, together with chemistry-matched analog systems where needed. As reflected in the system schematic shown in Figure 1, the design focuses on the wastewater-to-receiving-water continuum relevant to municipal outfalls and related coastal discharge environments. A key feature of the project is that characterization will begin within the plant context itself. Following an initial setup period, the study will establish baseline wastewater chemistry before any $Ca(OH)_2$ perturbation, including characterization of relevant plant streams where access allows and final effluent before discharge. These measurements will be paired with recurring receiving-water collections outside the plant so that wastewater-side chemistry controls can be distinguished from receiving-water and dilution effects.

In practice, the study will focus primarily on final treated effluent and the receiving waters into which it would mix after discharge. Limited characterization of upstream or intermediate plant chemistry may be included where useful to interpret why treated effluent differs across plant types or operating conditions. This is important because treatment-stage differences, ammonium oxidation status, residual organic matter, and nutrient transformations may influence the carbonate chemistry of the final effluent and therefore its response to $Ca(OH)_2$ addition. The goal is not to fully characterize treatment processes, but to provide enough plant-side context to explain observed differences in precipitation risk and retained alkalinity across wastewater discharge settings.

This project is intentionally designed as an ex-situ, deployment-oriented experimental study rather than a field dosing pilot. Its experimental framing is guided by our broad knowledge of wastewater treatment plant chemistry and receiving water chemistry on a global scale and supplemented by past and on-going synthesis efforts of our own and others[3,5,6]. The work is therefore best described as bench-scale and laboratory-based perturbation testing using authentic wastewater and receiving-water matrices. That choice reduces permitting and interpretation risk while still preserving real-world relevance. It also allows the project to isolate the chemistry and mixing controls that matter most before moving into more expensive and logistically complex pilot work. In that sense, the approach is best-in-class for the specific bottleneck being addressed: it combines realistic environmental matrices, controlled experimental conditions, and direct translation to future deployment and MRV questions.

Measured variables will include, but not be limited to, the core carbonate system parameters needed to quantify alkalinity persistence and precipitation behavior, including TA, DIC, pH, pCO_2, salinity, and temperature. Where useful, additional contextual measurements associated with wastewater influence, such as dissolved oxygen, turbidity, ammonium, nutrient status, or organic-matter-linked indicators, will be incorporated to help explain departures from idealized seawater behavior. Controlled $Ca(OH)_2$ perturbation experiments will then be used to test how different delivery modes, starting chemistries, and dilution pathways influence retained alkalinity and the onset of secondary precipitation under deployment-relevant conditions.

Experimental results will be interpreted both within and across regions. Within-region comparisons will be used to identify how plant type, treatment performance, and receiving-water setting influence $Ca(OH)_2$ behavior under local conditions. Cross-region comparisons will then be used to determine which relationships appear robust across contrasting climatic, hydrologic, and wastewater-chemistry settings, including differences in wastewater DIC and receiving-water TA/DIC associated with treatment context and freshwater influence, and which are more context-specific. This comparative design is especially important for future deployment because it allows the project to generate constraints that are transferable enough to support comparative evaluation and pre-pilot design, while still identifying when additional context-specific characterization would be required.

This project is differentiated from simplified laboratory studies by its emphasis on realistic wastewater matrices and receiving waters, and from site-specific pilots by its focus on extracting transferable constraints rather than demonstrating a specific field deployment. The expected outputs fall into four categories: a validated experimental dataset on precipitation behavior and alkalinity retention; empirical interpretation linking chemistry and mixing conditions to short-timescale performance; technical interpretation relevant to future trial evaluation, together with a chemistry-informed technical summary for evaluating retained alkalinity and precipitation risk; and MRV-relevant measurement priorities for interpreting alkalinity persistence in open systems. Together, these outputs are intended to improve pre-pilot design, strengthen comparative evaluation of candidate receiving-water environments, sharpen monitoring strategy, and provide more defensible technical foundations for wastewater-enabled and coastal alkalinity enhancement.

References

4. Cao, X., Chen, S., Liu, Y., Long, G. & Xu, Y. J. Domestic wastewater is an overlooked source and quantity in global river dissolved carbon. Nat. Commun. 16, 7522 (2025).
5. Pan, Y. et al. The role of Mg^{2+} in inhibiting $CaCO_3$ precipitation from seawater. Mar. Chem. 237, 104036 (2021).
6. Zheng, L., Hu, Y., Su, B., Chen, Q. & Liu, J. The potential of wastewater treatment on carbon storage through ocean alkalinity enhancement. Sci. Adv. 11, (2025).

d. **Risks:** What are the biggest risks to your project's success and how will you mitigate those? Include scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. <500 words

The primary scientific risk is that precipitation behavior may vary enough across wastewater and receiving-water conditions to limit how broadly results can be generalized. This will be mitigated by designing the study around representative chemistry classes rather than a single site or water type, informed by our collective understanding of wastewater treatment plant chemistry and receiving-water chemistry across a range of settings and by past and ongoing synthesis efforts from our team and the wider field.

A second scientific and technical risk is that controlled experiments will not fully capture real plume hydrodynamics and near-field mixing behavior. The project addresses this by focusing on realistic dilution ranges, discharge-relevant mixing conditions, and comparative water classes, while keeping the scope ex situ so that core chemistry questions can be resolved before field validation.

A third risk is that wastewater-associated biogeochemical variability, including ammonium oxidation, nutrient loading, and organic matter respiration, may introduce greater heterogeneity than expected. This will be addressed by explicitly incorporating wastewater-relevant chemistry classes into the experimental design rather than treating them as noise.

Execution risk is concentrated in multi-context sample access. To reduce this, the project is anchored in one primary temperate municipal wastewater and coastal receiving-water context, while comparator matrices from warm semi-enclosed coastal and/or tropical estuarine systems will be included only where access, timing, and partner approvals are feasible. If direct access to some environmental matrices is delayed, chemistry-matched analog systems will be used to preserve the comparative design.

Measurement uncertainty will be mitigated through established carbonate chemistry workflows, replicate experiments, and mass-balance-based interpretation of retained alkalinity. Ecosystem risk is limited because the proposed work does not include active in situ $Ca(OH)_2$ addition; instead, environmental relevance is maintained through ex situ experiments using authentic wastewater and receiving-water matrices. Because the proposed work is ex situ, environmental and permitting risks are much lower than for an active field trial.

A financial and project-management risk is that multi-context sample access, shipping, procurement, or permitting could increase costs or slow timelines. This will be mitigated by anchoring the core study in one primary wastewater and coastal receiving-water context, treating comparator matrices as modular additions rather than essential dependencies, and maintaining sufficient flexibility in sampling design and experimental matrices to preserve the project's main deliverables if a comparator workstream is delayed or removed from scope.

e. **Proposal to Frontier:** Please list your primary research deliverables, project timeline, and requested funding below. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables *(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes, or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)*	Validated experimental datasets describing $Ca(OH)_2$ addition outcomes across representative wastewater effluents, receiving waters, and mixing conditions. Empirical interpretation of precipitation behavior and alkalinity retention across representative chemistry classes and generalized dilution conditions. A chemistry-informed technical interpretation summary describing observed relationships among wastewater chemistry, receiving-water chemistry, dilution conditions, alkalinity retention, and precipitation behavior

	across the studied conditions. MRV-relevant interpretation guidance and parameterizations for near-field chemistry evolution in future wastewater-enabled and coastal OAE/IWAE trials.
Field impact *(Briefly describe how these deliverables resolve key bottlenecks for a specific CDR pathway or broader CDR field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)*	These deliverables will reduce a major uncertainty constraining OAE and IWAE scale-up: whether and under what conditions $Ca(OH)_2$ remains available as persistent alkalinity following addition in wastewater and coastal discharge settings. The results will strengthen pre-pilot design, improve comparative evaluation of candidate receiving-water environments, support more defensible dosing interpretation, and reduce uncertainty in open-system monitoring and net removal calculations.
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier generally prefers projects with clear readouts within ~18 months.)*	**Months 1–3** Complete project launch, finalize partner roles, define the representative chemistry envelope, and finalize the sampling, QA/QC, permissions, and shipment framework. During the winter period, complete procurement, mobilization, and analytical readiness activities needed to support the first seasonal field campaign. **Months 4–9** Conduct spring and summer plant-side and receiving-water characterization campaigns across the core geographies, complete rolling sample receipt and analysis, and generate chemistry-matched analog systems where needed. Use these data to initiate and refine controlled $Ca(OH)_2$ perturbation experiments. **Months 10–15**

	Complete autumn characterization and the core-controlled $Ca(OH)_2$ perturbation dataset, including analytical duplicates, replicate runs, confirmatory experiments, and archived observational and experimental records. **Months 16–18** Quantify precipitation thresholds and retained alkalinity relationships, complete uncertainty and MRV-relevant interpretation, and deliver the final technical synthesis, structured dataset package, and dissemination materials.
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	**$705,000**